Adil Faizi

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Education

York University

Bachelor of Engineering, Electrical Engineering (Honours) Sept 2023 - May 2027

- Course Work: Power Systems, Linear Control Systems, Applied Electromagnetism, Signal And Systems, Electric Circuit and Devices
- Co-op Eligibility: 4/8/12/16-month terms

Skills

Programming: MATLAB, Python, C, Java, Bash, Arduino
CAD & Circuit Design: AutoCAD, Revit, KiCad, ETAP
Analysis & Simulation: LTspice, MATLAB
Productivity: Microsoft Excel, Word, PowerPoint

Experience

General Clerk Habitat For Humanity – Brampton, ON June 2021 – August, 2022

- Collaborated with team members to inspect, test, and assess the safety and functionality of donated electrical and electronic equipment.
- Communicated effectively with supervisors and teammates to diagnose issues, perform basic repairs, and ensure proper documentation and safe handling while meeting operational deadlines.
- Demonstrated eagerness to learn by quickly adapting to new tools, procedures, and safety practices, gaining hands-on experience with wire strippers, crimpers, and voltmeters.

Projects

DSP Filter Design and Audio Restoration Project | **MATLAB**

- Designed a bandpass FIR filter in MATLAB meeting strict passband and stopband ripple specifications, computed filter length, coefficients, and frequency response.
- Analyzed corrupted audio using frequency-domain techniques to identify and isolate noise components.
- Implemented time- and frequency-domain noise removal methods and verified performance via ripple/attenuation metrics and listening tests.

Ultrasonic Distance Checker with I2C LCD Display | **Arduino**

- Developed a real-time distance measurement system using an ultrasonic sensor with live output on a 1602 I2C LCD.
- Programmed an Arduino Uno R4 Minima to acquire, process, and update distance readings at fixed intervals.
- Implemented efficient I2C communication to ensure smooth, responsive LCD updates.

Remote Controlled Car | Embedded C and Python

- Implemented an RC car system using microcontrollers and programming in Embedded C and Python, enabling wireless control and autonomous features
- Developed an object-oriented control framework to manage propulsion, steering, and sensor integration
- Integrated real-time messaging to log diagnostics such as speed, battery level, and location to a .txt file

Automated Irrigation System | **MATLAB and Java**

- Built a Java–MATLAB system that automates irrigation by detecting soil conditions and simulating large scale water-distribution logic.
- Integrated CO_2 and sunlight sensors, plus real-time mobile alerts via a custom messaging interface
- Optimized system performance through iterative testing to improve sensing accuracy and automated decision-making.